13. DNA dependent RNA polymerase catalyzes transcription on one strand of the DNA which is called the
    a) Alpha strand b) Anti strand
    c) Template strand d) Coding strand

14. Which of the following correctly represents the flow of genetic information?
    a) DNA → RNA → Protein
    b) RNA → DNA → Protein
    c) RNA → Protein → DNA
    d) Protein → RNA → DNA

15. Initiation codon is
    a) UUU b) UGA
    c) AUG d) UAG

16. A eukaryotic gene contains two kinds of base sequences which of these plays an important role in protein synthesis?
    a) Introns b) Exons
    c) Both a and b d) None of the above

17. Codon – anticodon interactions occur by
    a) Covalent bond
    b) Electrostatic interactions
    c) Hydrogen bonds
    d) Hydrophobic interaction

18. Which of the following RNA polymerases is responsible for the transcription of protein coding genes in eukaryotes?
    a) RNA Pol I b) RNA Poly II
    c) RNA Pol III d) RNA Pol IV

19. How are RNA molecules transported out of the nucleus
    a) Passive diffusion through the membrane
    b) Through membrane pores in an energy independent process
    c) Through membrane pores in an energy dependent process
    d) Through a channel in the membrane that leads to the endoplastic reticulation

20. During translation the codon in mRNA is actually "read" by
    a) The A site in the ribosomes
    b) The P site in the ribosomes
    c) The anticodon in a tRNA
    d) The anticodon is an amino acid

21. A complex of ribosome attached to a single strand of RNA is known as
    a) Polysome b) Polymer
    c) Polypeptide d) Okazaki fragment

22. Which of the following is the start codon
    a) AUG b) UGA
    d) UAA d) UAG

23. What is true about tRNA?
    a) It binds with an amino acid at its 3' end
    b) It has 5 double stranded regions
    c) It has a codon at one end which recognizes the anticodon of mRNA
    d) It looks like clover leaf in the three D structure

24. Which one of the following hydrolysis internal phosphodiester bonds in a polynucleotide chain?
    a) Lipase b) Exonuclease
    c) Endonuclease d) Protease

25. DNA element with ability to change position is called
    a) Cistron b) Transposon
    c) Intron d) Recon

26. Spliceosomes are not found in cells of
    a) Plants b) Fungi
    c) Animals d) Bacteria

27. During DNA replication Okazaki fragments are used to elongate
    a) The leading strand towards replication fork
    b) The lagging strand towards replication fork

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